Homework 2

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Instructions before the assignment 作业前须知:

- This assignment is for the **second week's homework** of the System Dynamics and Control course (hereafter referred to as SDC). The submission deadline is **before 11:59 PM on September 26, 2024**. Please ensure timely submission, as late submissions will result in point deductions. 本次作业为 SDC **第二周课程作业**，提交时间为 **2024/09/26 23:59** 前。
  - If submitted within 12 hours after the deadline, no points will be deducted (**this is allowed only twice, so focus on completing it properly rather than rushing to meet the deadline if you' re unable to finish on time**);
  - If submitted within three days after the deadline, **20% of the total score of this assignment will be deducted**;
  - If submitted within two week after the deadline, **30% of the total score of this assignment will be deducted**;
  - Any other late submissions will result in**a 40% deduction**;
- For late submissions, please contact the teaching assistant(TA), **Yan Jizhou**, via email at **yanjizhou312@163.com** or WeChat **y3139838452**. Make sure to use the filename format as "SDC-Assignment-x-Resubmission-Name-Student ID." 作业补交时通过邮箱或微信与助教闫济洲联系，邮件标题及文件名备注为"SDC-第 x 次作业补交-姓名-学号"。
- The assignment is graded out of 10 points. It consists of basic course questions (10 points) and extension questions (0 to 5 points). **The final assignment score for each submission will be the smaller value between 10 and the actual score**.Each assignment earns all points by completing only the basic assignment. 每次作业仅完成基础作业即可获得所有分数，鼓励同学们完成拓展课程题目。
- Students are encouraged to attempt the extension questions. Any extra points earned will be proportionally added to the final course grade for possible grade improvement.
- **Academic misconduct such as plagiarism is strictly prohibited in course work. Plagiarism and other academic misconduct will be dealt with in accordance with the relevant regulations of the University.**. If you have discussed the assignment with classmates or received help, please note at the beginning of the assignment: "This assignment was completed after discussion with XXX (classmate's name), and part of the thought process was referenced from XXX (classmate's name)."**Every contribution deserves recognition. Unacknowledged similarities between assignments will be treated as plagiarism.** **课程作业中严禁抄袭等学术不端行为。**如作业完成过程中经历了同学讨论/同学帮助的情况，请在作业最前面注明"本次作业经过与 XXX（同学姓名）讨论，参考了 XXX（同学姓名）的部分思路"。**每位同学的帮助都值得被记录。不被注明的作业雷同将被认为是抄袭行为。**

1. Problem 1

Use the Laplace Transform (LT) and its Inverse Laplace Transform (ILT) to solve the following differential equation.

$$\ddot{y}+2\dot{y}+y=2, y(0)=0, \dot{y}(0)=0$$

$$\ddot{y}+2\dot{y}+y=\sin t, y(0)=0, \dot{y}(0)=0$$

$$\ddot{y}-6\dot{y}+9y=0, y(0)=1, \dot{y}(0)=1$$

$$y^{(3)}-3\ddot{y}-4\dot{y}=0, y(0)=1, \dot{y}(0)=1, \dot{y}(0)=1$$

$$\ddot{y}+4\dot{y}+5y=\mathrm{e}^{-2x}\sin x, y(0)=0, \dot{y}(0)=0$$

$$\ddot{y}-4\dot{y}+4y=1+\sin x+x\mathrm{e}^{2x}, y(0)=0, \dot{y}(0)=0$$

$$x\ddot{y}+2\dot{y}=12\ln x, y(1)=0, \dot{y}(1)=0$$

2. Problem 2

Use the Laplace Transform (LT) and its Inverse Laplace Transform (ILT) to solve the following differential equation.You may retain the Laplace operator $\mathcal{L}$ and the inverse Laplace operator $\mathcal{L}^{-1}$ in the equation.

2.0 a

$$\tau\dot{y}+y=Ku(t)$$

Where τ and K are constants.

2.0 b

$$\frac{1}{\omega_n^2}\ddot{y}+\frac{2\zeta}{\omega_n}\dot{y}+y=Ku(t)$$

Where ω_n , ζ and K are constants.

2.0 c

Please decompose the obtained analytical solution into terms related only to the initial values and terms related to the external input $u(t)$.

2.0 d

We refer to the case where all initial states are zero and the input is a unit step signal as the zero-state response. Please provide the zero-state responses of the first-order system (a) and the second-order system (b).

We refer to the case where the external input is zero and the initial states are non-zero as the zero-input response. Please provide the zero-input responses of the first-order system (a) and the second-order system (b).

3. Problem 3

In this problem, we will briefly explore the similarities and differences between the state-space form and the input-output representation using Laplace transforms in the solution process.

3.1 a(Input/Output Representation)

Consider a system represented as:

$$\left[\left(\frac{\mathrm{d}}{\mathrm{d}t}+1\right)^2\left(\frac{\mathrm{d}}{\mathrm{d}t}+2\right)\right]y=1,\quad y(0)=0,\quad \dot{y}(0)=0,\quad \ddot{y}(0)=0$$

where $\frac{\mathrm{d}}{\mathrm{d}t}$ is the differential operator. This differential equation is equivalent to:

$$y^{(3)}+4\ddot{y}+5\dot{y}+2y=1$$

Please use the Laplace Transform and its inverse to solve this differential equation.

3.2 b(State-Space Representation)

Consider the same system as in part a, whose state-space representation is:

$$\begin{cases} x_1 = y \\ x_2 = \dot{y} \\ x_3 = \ddot{y} \end{cases}$$

Let $X=[x_1,x_2,x_3]^T$, $Y=[y]$, and $U=[u]$.
Thus, it can be written in the form of a system of linear differential equations:

$$\begin{cases} \dot{X} = AX + BU \\ Y = CX + DU \end{cases}$$

The key to solving this system of linear differential equations lies in solving the equation $\dot{X}=AX+BU$. This is a first-order linear ordinary differential equation system, and common methods for solving it include:

1. Decompose the matrix A into $A=P^{-1}JP$, where J is the Jordan canonical form and P is the matrix of generalized eigenvectors.
2. Make the transformation $Z=PX$, then the original equation becomes $\dot{Z}=JZ+(PB)U$.
3. Solve by independently calculating variables or using the matrix exponential method.

In this problem, you are not required to solve this system of linear differential equations. The tasks you need to complete in this question are:

1. Based on the system differential equation in 3.a, find the coefficient matrices A, B, C, and D in the state-space representation.

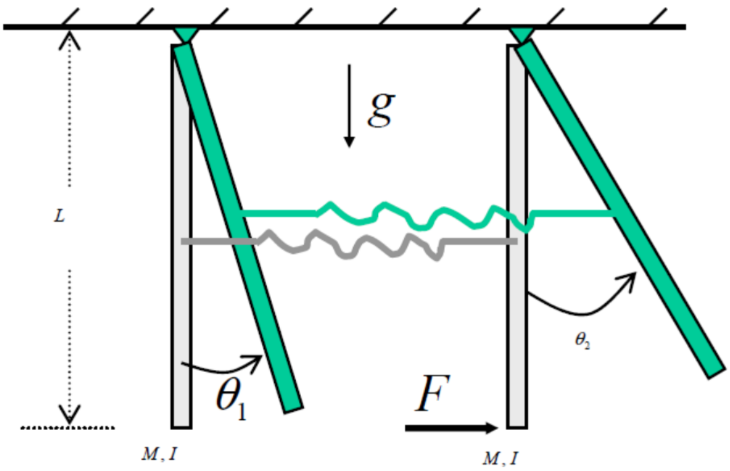
2. Decompose the matrix A into its Jordan canonical form; you do not need to find the matrix P. (This process is rather cumbersome and you may use MATLAB, Mathematica, or other tools to complete it.)

3. Compare the eigenvalues in the Jordan canonical form with the solutions of the differential equation in 3.a. What do you observe?

3.2 c Simple Application

This problem does not restrict the solving method, but you might consider how to express the answer more concisely.

Consider the double pendulum shown in the figure:



where the two pendulums have equal mass M and moment of inertia I (with respect to the joint) and are connected at the center through a spring with a stiffness of K and a damping coefficient of B_s. The spring is un-stretched when the two pendulums are at the downward positions. At time zero, a horizontal force F is applied to the tip of the second pendulum as shown. Model the system through the following procedure:

Assume all angles are small, which means you can make the following approximations: $\sin(\theta) \approx \theta$, $\cos(\theta) \approx 1$. All initial states are zero.

Attempt to establish differential equations and solve for the variation of θ_1 over time. The order of the differential equation is relatively high; you may retain the Laplace operator $\mathcal{L}$ and the inverse Laplace operator $\mathcal{L}^{-1}$ in the final expression.

Tips: There are three ways of thinking for your reference, but you can also propose your own method.

1. List all differential equations and simplify them into the form where F is the input and θ_1 is the output. Then use the Laplace transform to solve.

2. (Transfer Function) List all differential equations, use the Laplace transform to convert the differential equations into algebraic equations (transfer function form), solve them simultaneously, and then perform the inverse transform.

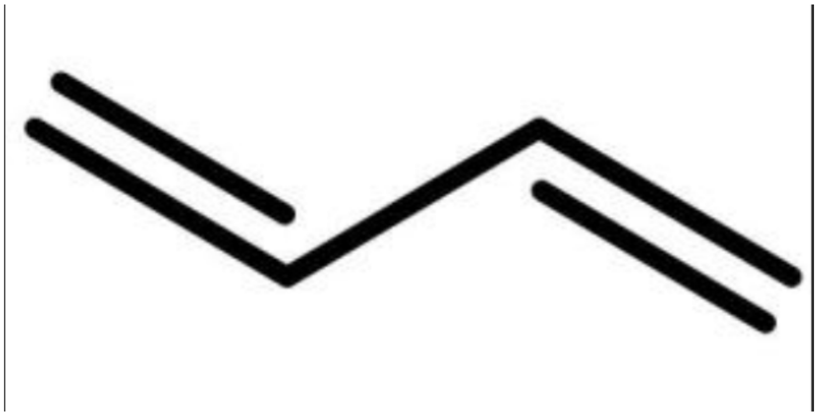
3. Treat the entire system as a two-degree-of-freedom system, write matrix equations by considering it as a whole, and solve the matrix equations using the Laplace transform.

4. *Problem 4: Molecular Orbitals and State Space (breadth extension)

The molecular orbitals of complex organic molecules can be obtained by solving the Schrödinger equation. We can express the state of an electron as a superposition of probability amplitudes in various states. Below, we will attempt to represent the electron's state in molecular orbitals in the form of state space and try to solve the Schrödinger equation for organic molecules.

In this problem, you may try to directly solve the differential equations in state space or attempt to convert the matrix equations into higher-order linear equations for solving.

4.1 a 1,3-Butadiene



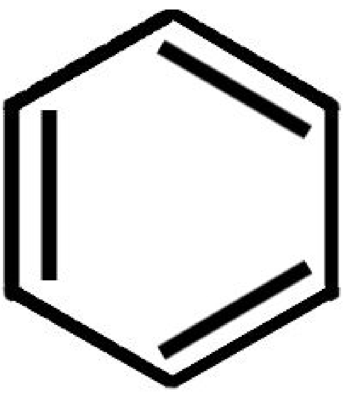
The figure shows the skeletal formula of 1,3-butadiene. In 1,3-butadiene, suppose an additional electron can occupy the sp^3 hybrid orbital of each carbon atom. Assume that the energy of the electron on each carbon atom is E_0, and the energy barrier to transition to a neighboring carbon atom is A. Therefore, we can write the Hamiltonian equation as:

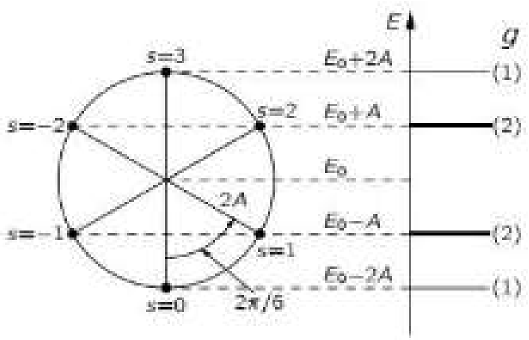
$$
\mathrm{i}\hbar\frac{\mathrm{d}C}{\mathrm{d}t} = \begin{bmatrix} E_0 & -A & \cdot & \cdot \\ -A & E_0 & -A & \cdot \\ \cdot & -A & E_0 & -A \\ \cdot & \cdot & -A & E_0 \end{bmatrix} C
$$

Where $C = [C_1, C_2, C_3, C_4]^T$ represents the probability amplitudes of the electron on each carbon atom. Please try to solve this equation.

Tips: In this problem, the coefficient matrix of C is a semi-circulant matrix with eigenvalues $E_0 - 2A$, $E_0 - A$, $E_0 + A$, and $E_0 + 2A$.

4.2 Benzene





The benzene has a Π_6^6 delocalized large π bond, but we can still make similar assumptions as in the previous question, giving its Hamiltonian equation as:

$$
\mathrm{i}\hbar\frac{\mathrm{d}C}{\mathrm{d}t}=\begin{bmatrix} E_0 & -A & \cdot & \cdot & \cdot & -A \\ -A & E_0 & -A & \cdot & \cdot & \cdot \\ \cdot & -A & E_0 & -A & \cdot & \cdot \\ \cdot & \cdot & -A & E_0 & -A & \cdot \\ \cdot & \cdot & \cdot & -A & E_0 & -A \\ -A & \cdot & \cdot & \cdot & -A & E_0 \end{bmatrix} C
$$

Where $C=[C_1,C_2,C_3,C_4,C_5,C_6]^T$ represents the probability amplitudes of the electron on each carbon atom. Please try to solve this equation.

It is worth mentioning that the Hamiltonian matrix of the benzene is a circulant matrix with eigenvalues $E_0-2A, E_0-A, E_0+A, E_0+2A, E_0+A, E_0-A$. The eigenvalues of the Hamiltonian matrix are exactly the eigenenergy levels of the benzene. The multiplicity of an eigenvalue indicates how many electrons can be filled at that energy level. These are among the theories that have been experimentally verified.